🔬 SCIENTISTS AND INVENTIONS — FULL MIND MAP

वैज्ञानिक एवं आविष्कार · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📝 Topics in this chapter:

1. Early Instruments, Optics & Foundational Inventions

- 1718 me, James Puckle naam ka ek English inventor ne Defence Gun invent kiya jo mechanical machine gun ke development me important steps me se ek prove hua. unhone Defence Gun ka patent karayaa, jo us time tak paaee jane wali manual roop se operated revolving machine gun ke sabse refined design ko represent karti hai. *[Q1]*
- Thermometer ya thermoscope ki discovery famous Italian physicist Galileo Galilei se associated hai. Practical refracting telescope ko Galileo ne 1609 me major improve karke astronomical observations ke liye use kiya. Microscope ek aisa device hai jo bahut chhoti nearby objects ko dekhane me help karta hai jinhen naked eyes se clearly nahi dekha ja sakta hai. *[Q2–Q4]*
- Typewriter ka invention 1867 me Christopher Latham Sholes ne kiya tha (A-2). ek German mechanical engineer aur physicist Wilhelm Conrad Roentgen ne 1895 me X-ray ya Roentgen rays ke roop me jaanee jane wali wavelength range me electromagnetic radiation discover ki. unhone effective roop se X-Ray Machine invent kiya. British physicist aur chemist Sir William Henry Bragg ne 1915 me apne son William Lawrence Bragg ke saath physics me Nobel Prize share kiya tha. Johannes Gutenberg ne printing press invent kiya (A-2). *[Q5–Q8]*

> ⚠️ EXAM TRAP — Q3 — 'Telescope invented by Galileo' exam shorthand hai
>
> ► Galileo ne first telescope invent nahi kiya tha. Hans Lippershey ko 1608 ke telescope patent application ka common credit milta hai; Galileo ne 1609 me telescope ko kaafi improve karke astronomy me famous kiya.
> ► Source answer/coverage preserve hai, lekin concept ke liye yaad rakhein: Lippershey → early telescope; Galileo → major improvement & astronomical use.

2. Engines, Communication & Everyday Technology

- Steam engine ka development 17th century ke end me start hua (Thomas Savery aur thomas Newcomen ne early version banaye the). Haalaanki, 1763 me Scottish engineer aur inventor James Watt dwara sabse important improvements kiye gaye jisne steam engine ko industrial use ke liye practical bana diya. Isliye, unhe widely iska inventor maana jata hai. *[Q9–Q10]*
- Thomas Alva Edison ko practical, long-lasting incandescent electric lamp ke liye widely jaana jata hai. Modern gas engine (Gas engine) 1885 me German engineer Gottlieb Daimler dwara develop kiya gaya tha. (Rudolf Diesel ne 1895 me Diesel engine developed kiya tha). Lewis Edson Waterman ne 1883 me New York me capillary-feed fountain pen invent kiya. unhone 1884 me pehle practical fountain pen ka patent karaya tha. *[Q11–Q13]*
- Scottish physicist Robert Watson-Watt radar (Radio Detection and Ranging) ke development me pioneer aur important contributor the. Swedish Scientist aur innovator Alfred Nobel ne Dynamite invent kiya. Famous Nobel Foundation ki establishment 1900 me unki will ke basis par ki gayi thi. Mechanical Television ka invention John Logie Baird (J.L. Baird) dwara 1926 me Britain me kiya gaya tha. Scottish engineer J.L. Baird ka naam television ke invention se linked hai. *[Q14–Q17]*
- Theodore Maiman ne LASER (Light Amplification by the stimulated emission of Radiation) invent kiya tha. 16 maee, 1960 ko unhone California ke maaleeboo me Hughes Research Laboratories me apni laboratory me ruby kristal se pehle varking LASER ka successfully demonstration kiya tha. *[Q18]*

⚠ EXAM TRAP — Q9, Q11 & Q12 — Inventor attribution me nuance

▸ Steam engine Savery aur Newcomen ke work se evolve hua; James Watt ne major efficiency improvements kiye. Exam books me often simply 'James Watt—steam engine' diya hota hai.
▸ Incandescent lighting me multiple contributors the; Edison commercially practical aur durable incandescent-lamp system ke liye best known hain.
▸ Four-stroke internal-combustion/gas engine Nikolaus Otto (1876) se strongly associated hai; Daimler ne baad me high-speed petroleum engines develop kiye. Q12 ko source-specific wording ke context me padhein.

3. Penicillin, X-rays & Discovery Matching

- Penicillin ki discovery 1928 me Scottish Scientist Alexander Fleming ne ki thi, je. Parkins ne nahi. Baakee jode correctly matched hain. Pehla sachcha enteebaayotika, Penicillin, Alexander Fleming dwara discover kiya gaya tha. Alexander Fleming ne 'Penicillium notatum' (Penicillium notatum) molda (fungus) se antibiotic Penicillin ki discover ki thi. Sir Alexander Fleming ne Penicillin ki discover ki thi, jiske liye unhe Howard Florey aur Ernst Boris Chain ke saath 1945 me Physiology ya Medicine me Nobel Prize milaa. *[Q19–Q23]*
- Radium ki discovery 1898 me piyare kyooree aur maidam kyooree ne ki thi (A-2). Penicillin ki discovery 1928 me Alexander Fleming ne ki thi (B-1). X-ray ki discovery 1895 me German physicist Wilhelm Conrad Roentgen ne ki thi. *[Q24–Q25]*
- Johannes embros Fleming ne 1904 me pehle varking daayod invent kiya. (B) Traanjistar ka invention 1947 me American physicalvidon Johannes baardeena, vaaltar braitan aur William shokle ne kiya tha. (A) *[Q26]*
- Alfred Nobel ek Swedish chemist the jinhone Dynamite invent kiya tha (A-2). Alexander Fleming ek Scottish jeevavijnyaanee the jinhone 1928 me antibiotic 'Penicillin' ki discover ki thi (B-3). *[Q27]*

⚠ EXAM TRAP — Q24 — Jenner: Smallpox, Measles nahi

▸ Edward Jenner ne 1796 me SMALLPOX ke against vaccination pioneer ki. 'Measles/खसरा' aur 'smallpox/चेचक' alag diseases hain; source wording inhe mix karti hai.
▸ Exam ke liye safe fact: Edward Jenner → smallpox vaccine. 'Measles vaccine—Jenner' memorize na karein.

4. S. Chandrasekhar & Stellar Astrophysics

- Indian-American khagol Scientist subrahmanyan chandrashekhar ko 'chandrashekhara seemaa' (Chandrasekhar Limit) ke principle ke liye jana jata hai. Subrahmanyan chandrashekhar ko stars ki structure aur development ke liye important physical processon ke unke saiddhaantik study ke liye 1983 me physics ka Nobel Prize diya gaya tha, jo ki Astrophysics ke field me aataa hai. S. Chandrashekhar ne prove kiya ki jina stars ka mass Sun ke mass ke 1.44 gunaa (chandrashekhara seemaa) se kam hota hai, ve apanaa nuclear fuel samaapt hone par sthira white vaaman (white dwarfs) bana jate hain. Yadi ve is mass se zyada ho jate hain, to ve aur zyada dhaha jate hain. *[Q28–Q30]*

⚠ EXAM TRAP — Q28 — Chandrasekhar Limit ≠ original 'black-hole principle'

▸ Chandrasekhar ka key contribution stable white dwarfs ki mass limit aur uske baad stellar-collapse implications hain. Black-hole theory kai physicists ke work se develop hui, including Schwarzschild aur later relativists.
▸ Agar source exact legacy question poochhe to intended answer S. Chandrasekhar hai; conceptually Chandrasekhar limit yaad rakhein.

5. Photoelectric Effect, Einstein & Relativity

- Photoelectric effect wo ghatana hai jisme electromagnetic radiation ko avashoshit karne par kisi material se ya uske andar se electrical charged particle nikalate hain. Scientist Albert Einstein Plank ke kvaantam principle ke basis par photoelectric effect ki sarala aur tathyatmak vyakhy dene ke liye famous hain. Einstein ko "Theoretical Physics me unki sevaaon aur specially photoelectric effect ke niyama ki unki discovery ke liye" 1921 me physics ka Nobel Prize diya gaya tha. Jaisaa ki established hai, Albert Einstein ko mainly photoelectric effect ke unke niyama (law of the photoelectric effect) ke liye Nobel Prize se sammaanit kiya gaya tha. *[Q31–Q36]*
- 'Theory of relativity' me usually Albert Einstein ke do paraspar related principle shamil hote hain: vishesh relativity (1905 me prastaavita) aur general relativity (1915 me prakaashita). Vishesh relativity gravity ke abhaav me sabhi physical ghatanaaon par laagoo hoti hai, ye tarka dete hue ki space aur time atoota roop se jude hue hain. General relativity gravity ke niyama aur prakriti ke balon ke saath iske sanbandh ki vyakhy karti hai. *[Q37–Q38]*
- Physics me apne groundbreaking kaam ke alawa, Albert Einstein Violin aur piyano bajaane me bhi kushal the. Sangeet unke jeevan ka ek gahara aur important hissaa tha. Chauth dimensions (fourth dimension) usually teen sthanik aayamon (lanbaaee, chaudaaee aur depression) ke saath ek other dimensions ke roop me 'time' ko refer karta hai. *[Q39–Q40]*

⚠ EXAM TRAP — Q40 — Four-dimensional spacetime attribution

► Einstein ki special relativity ne space aur time ko physically unite kiya, lekin Hermann Minkowski ne 1907–08 me famous mathematical four-dimensional spacetime formulation diya.
► Source-style MCQ Einstein expect kar sakta hai; deeper physics me Einstein ki relativity aur Minkowski spacetime formalism ko distinguish karein.

6. C.V. Raman & National Science Day

- 1930 me, Sir C.V. Raman light ke scattering (scattering) aur ramana prabhaav ki discovery par apne kaam ke liye physics me Nobel Prize mila karne wale pehle eshiyaee aur Indian bane. unhone ye discovery February, 28, 1928 ko ki thi, jise India me National science divasa ke roop me manaay jata hai. Jaisaa ki established hai, do C.V. Raman ko varsha 1930 me physics me Nobel Prize se sammaanit kiya gaya tha. *[Q41–Q42]*
- Indian physicist Sir C. V. Raman dwara February, 28 1928 ko ramana prabhaav ki discovery ko chihnit karne ke liye India me hara saala February, 28 ko National science divasa manaay jata hai. India me National science divasa February, 28 ko manaay jata hai. Jaisaa ki established hai, ramana prabhaav ki discovery ke upalakshy me India February, 28 ko National science divasa manaata hai. Ye question technical roop se incorrect hai. National science divasa (February, 28) C.V. Raman ke janmadin par nahi, balki ramana prabhaav ki discovery ke upalakshy me manaay jata hai. *[Q43–Q46]*
- C.V. Raman ka janma 7 navanbara, 1888 ko hua tha. Isliye, unki janma shataabdee (100veen varshagaantha) varsha 1988 me manaaee gayi thi. Sandarbh ke liye: Charles Darwin ka janma 1809 me hua tha, Ramanujan ka janma 1887 me hua tha, aur Einstein ka vishesh Theory of relativity 1905 me prastaavit kiya gaya tha. *[Q47]*

⚠ EXAM TRAP — Q46 — National Science Day birthday celebration nahi hai

► 28 February C.V. Raman ke Raman Effect discovery (28 February 1928) ko mark karta hai. Raman ka birth 7 November 1888 ko hua tha.
► Isliye National Science Day ko Raman ke birthday se jodne wala question flawed premise hai.

7. Indian Scientists, Scientific Fields & Contributions

- 'da maina hoo nyoo inphinitee' 2015 ki British baayograaphikal draama philma hai jo 1991 ki isi naam ki kitaab par based hai. Indian khagol Scientist je.vee. Naarleekar ne 'relativity ka new principle' pratipaadit kiya. Varishth Scientist phreda hoyala ke sahayog se, unhone gravity ka hoyala-naarleekara principle developed kiya, jisme macha ke principle ko shamil karke Einstein ke relativity ke principle me sanshodhan karne ki maanga ki gayi thi. Indian khagol Scientist jayant vishnu (je.vee.) naarleekar ke according, Astrology (astrology) ek chhadm science hai aur ise ek sachcha science nahi maana jata hai, kyonki isme anubhavajany Scientist tareekon dwara satyapit poorvaanumaan power ka abhaav hai. *[Q48–Q50]*
- Do. Homee jahaangeer bhaabha ek Indian nuclear physicist the, jinhone India me Nuclear Energy programme ke development me highly important bhoomika nibhaaee. Unhe universally "Indian nuclear programme ke janak" ke roop me jana jata hai. Choonki homee jahaangeer bhaabha ko India ke Nuclear Energy programme ke janak maana jata hai, isliye unki smriti me mainly Nuclear Energy ke field me asaadhaaran yogadaan ke liye Scientiston ko 'Homi Bhabha award' diya jata hai. *[Q51–Q52]*
- Steephan William hoking (1942 - 2018) world famous British saiddhaantik physicist, brahmaand scientist aur author the. Unhe ek shaanadaar Scientist ke roop me widely maana jata hai, jinhen black hole, relativity aur 'e breeph histree opha taaima' jaisee lokapriy science pustakon par unke kaam ke liye jana jata hai. Sir Jagdish Chandra Bose ek bahushrut (polymath) the: ek prakhyat Indian physicist, jeevavijnyaanee, baayophijisista, vanaspatishaastree aur science katha ke early lekhaka. *[Q53–Q54]*
- Do. Raja Ramanna nuclear physics ke liye jane jate the (A-2). Do ema.esa. Svaameenaathan ek famous Indian krishi Scientist (India me harita kraanti ke janak) hain (B-5). *[Q55–Q56]*
- Ek shaanadaar Indian physicist do Raja Ramanna ne India ke nuclear bama ke development me highly yogadaan diya. Unki pratyaksh dekharekh aur netritv me, India ka pehla nuclear test (codenema 'smaailinga buddhaa') 1974 me successfully kiya gaya tha. Do Har Govind Khorana ko 1968 me chikitsa me Nobel Prize se sammaanit kiya gaya tha. C.V. Raman ne 1930 me physics me Nobel Prize jeetaa. S. Chandrashekhar ne 1983 me physics me Nobel Prize jeetaa. *[Q57–Q58]*

⚠ EXAM TRAP — Q55 — Matching list me source/option mismatch

► M.S. Swaminathan ka specialized field agricultural science/genetics aur Green Revolution hai, ‘plant chemistry’ nahi. Agar option list me agriculture hi absent ho to item defective hai.
► Safe mapping: Raja Ramanna—nuclear physics; M.S. Swaminathan—agricultural science; U.R. Rao—space research; Meghnad Saha—astrophysics/thermal ionization. In pairings ko isi tarah yaad rakhen.

8. Recent Science Prizes in the Source

- Physics me 2023 ka Nobel Prize piyare egostinee, pherenk kroja aur ainee ela'huiliyara ko joint roop se "praayogika tareekon ke liye diya gaya jo matter me ilektron speedsheelata ke study ke liye light ke etosekand spandan utpann karte hain." Option (B) 2022 Nobel Prize ka description karta hai, aur option (C) physics me 2021 ke Nobel Prize ka description karta hai. *[Q59]*
- Saankhyikee me 2023 ka international award, jise aksar saankhyikee ke 'Nobel Prize' ke same maana jata hai, ek highly sammaanit Indian-American ganitajny aur saankhyikeevid Calyampudi Radhakrishna Rao (C.R. Rao) ko saankhyikee ke field me unke smaarakeey yogadaan ko recognition dene ke liye provide kiya gaya tha. *[Q60]*

✓ Source coverage: Q1–Q60